

On Edge-graceful Spectra of Square of Paths with Odd Order

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Abstract

For a simple path P_r on r vertices, the square of P_r is the graph P_r^2 on the same set of vertices of P_r , and where every pair of vertices of distance 2 or less in P_r are connected by an edge. Given a (p, q) -graph G with p vertices and q edges, and an integer $k \geq 0$, G is said to be k -edge-graceful if the edges can be labeled by $k, k+1, \dots, k+q-1$ so that the induced vertex sums (mod p) are distinct, where the vertex sum at a vertex is the sum of the labels of all edges incident to such vertex. We call the set of all such k the edge-graceful spectrum of G , and denote it by $egI(G)$. In this article, the edge-graceful spectrum $egI(P_r^2)$ for square of paths is completely determined for odd r .

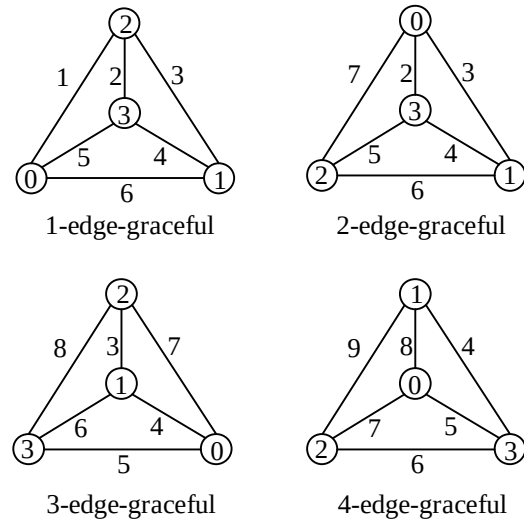


Figure 1: Various k -edge-graceful-ness of K_4

1 Introduction

Given an integer $k \geq 0$, a graph $G = (V, E)$ with p vertices and q edges is said to be k -edge-graceful if there is a bijection $f : E \rightarrow \{k, k+1, k+2, \dots, k+q-1\}$ such that the induced vertex sum $f^+ : V \rightarrow \mathbb{Z}_p$ where $f^+(u) = \sum \{f(u, v) : (u, v) \in E\} \pmod{p}$ is also a bijection. Such labeling is called a k -edge-graceful labeling. The study of the class of 1-edge-graceful graphs was initiated by S.P. Lo [18], and such class is also called edge-graceful graphs. Edge-graceful labeling can be viewed as a dual concept of the well-known graceful labeling. A good account on graceful labeling problems and other types of graph labeling problems can be found in the dynamic survey of J.A. Gallian [2]. Figure 1 shows that K_4 is k -edge-graceful for $k = 1, 2, 3, 4$.

We have the following necessary condition for being a k -edge-graceful graph, which is a generalization of Lo's condition in [18]: if a (p, q) -graph G is k -edge-graceful then it satisfies the condition that $q(q+2k-1) \equiv \frac{p(p-1)}{2} \pmod{p}$. A conjecture similar to the well-known graceful tree conjecture is that the above mentioned necessary condition is also sufficient for 1-edge-graceful graphs, particularly restricted to the class of trees.[7] The theory of 1-edge-graceful graphs is completely different from that of other k -edge-graceful graphs. For example, trees of order 4 are 2-edge-graceful but not 1-edge-graceful. (please see Figure 2) Some classes of 1-edge-graceful graphs are investigated in [1], [3]-[14], and [17]-[25]. Some properties of k -edge-graceful graphs are considered in [4, 9, 18].

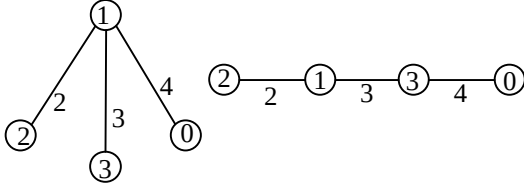


Figure 2: Trees are 2-edge-graceful but not 1-edge-graceful

We denote the set of all integers $k > 0$ such that G is k -edge-graceful by $egI(G)$. This set is called the edge-graceful spectrum of G . In [4, 9], the edge-graceful spectra of wheels and some $(p, p+1)$ -graphs are studied.

Notice that since we are considering the concept modulo the number of vertices p , therefore it is clear the calculation of the edge-graceful spectrum of graphs can be reduced to a finite set. Let us state the fact as a lemma which we will use frequently later:

Lemma 1. *If a (p, q) -graph G has a non-empty edge-graceful spectrum, and k is in $egI(G)$, then $k + t \cdot p$ is in $egI(G)$ for integers t .*

For a graph G , the square of G is the graph G^2 on the same set of vertices of G , and where every pair of vertices of distance 2 or less in G are connected by an edge. For the sake of convenience, the graph of P_r^2 can be drawn as an edge disjoint union of a inner path with an outer cycle. (please see Figure 3 for an example)

The purpose of this article is to determine the edge-graceful spectrum of the square of paths. Let us calculate the necessary conditions for being k -edge-graceful for the square of paths first:

Theorem 2. *If the graph P_r^2 is k -edge-graceful, then*

1. *Whenever r is odd,*
 - (a) *If $3 \mid r$, then $k \equiv 2 \pmod{\frac{r}{3}}$;*
 - (b) *If $3 \nmid r$, then $k \equiv 2 \pmod{r}$.*
2. *Whenever r is even,*
 - (a) *If $r \equiv 0 \pmod{4}$,*
 - i. *If moreover $3 \mid r$, then $k \equiv \frac{r}{12} + 2 \pmod{\frac{r}{6}}$;*
 - ii. *If $3 \nmid r$, then $k \equiv \frac{r}{4} + 2 \pmod{\frac{r}{2}}$.*
 - (b) *If $r \equiv 2 \pmod{4}$, then $egI(P_r^2) = \emptyset$*

Proof. The necessary condition of being a k -edge-graceful graph is $q(q+2k-1) \equiv \frac{p(p-1)}{2} \pmod{p}$. Note that the graph P_r^2 has r vertices and $2r-3$ edges. Therefore, the necessary condition becomes $-3(2k-4) \equiv r(r-1)/2 \pmod{r}$, and the calculations are as follows:

1. r is odd. Then the necessary condition becomes $3k \equiv 6 \pmod{r}$.
 - (a) If $3 \mid r$, then $k \equiv 2 \pmod{\frac{r}{3}}$.
 - (b) If $3 \nmid r$, then $k \equiv 2 \pmod{r}$.
2. r is even. Then the necessary condition becomes $6k \equiv \frac{r}{2} + 12 \pmod{r}$.
 - (a) If $r \equiv 0 \pmod{4}$, i.e. $r = 4n$ for $n \in \mathbb{Z}$, then the necessary condition is $3k \equiv n + 6 \pmod{2n}$.
 - i. If $3 \mid r$, then $k \equiv \frac{n}{3} + 2 \pmod{\frac{2n}{3}}$.
That is $k \equiv \frac{r}{12} + 2 \pmod{\frac{r}{6}}$.
 - ii. If $3 \nmid r$, then $k \equiv n + 2 \pmod{2n}$.
That is $k \equiv \frac{r}{4} + 2 \pmod{\frac{r}{2}}$.
 - (b) If $r \equiv 2 \pmod{4}$, i.e. $r = 4n + 2$ for $n \in \mathbb{Z}$. Then the necessary condition becomes $6k \equiv 2n + 13 \pmod{4n + 2}$. But this equation is incompatible.

□

2 Main Results

We would like to show that the necessary condition in Theorem 2 is also sufficient for the graph P_r^2 . By Lemma 1 and Theorem 2, in order to determine the edge-graceful spectrum of P_r^2 for odd r , it can be reduced into the following cases:

1. Whenever $3 \nmid r$, to show that it is 2-edge-graceful;
2. Whenever $3 \mid r$, to show that it is 2-edge-graceful, $(n+2)$ -edge-graceful, and $(2n+2)$ -edge-graceful, where $n = \frac{r}{3}$.

Combining the above cases, we will show that P_r^2 is 2-edge-graceful, where r is odd, through proving the following several theorems. First of all, we show that:

Theorem 3. *P_r^2 is 2-edge-graceful, where r is odd.*

Proof. Let the path P_r be the graph with vertex set $\{v_1, v_2, \dots, v_r\}$ and the edge set $\{v_i v_{i+1} : 1 \leq i \leq r-1\}$. P_r^2 is a graph on the same set of vertices

of P_r and where any pair of vertices of distance 2 or less are connected by an edge. Assume $e_{i,j}$ denotes the edge $v_i v_j \in E(P_r^2)$. Then the outer edges $e_{1,2}, e_{r,r-1}$, and $e_{i,i+2}$ where $1 \leq i \leq r-2$, form an odd cycle, and the inner edges $e_{i,i+1}$ where $2 \leq i \leq r-2$ form an even path (please see Figure 3 as an example of P_7^2). Next we will label the edges of the outer cycle and the edges of the inner path individually. To show that the graph P_r^2 is 2-edge-

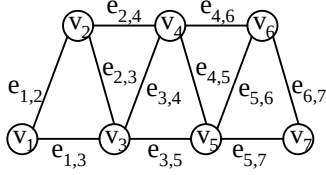


Figure 3: The index of vertices in P_7^2

graceful, we must label the integers $2, 3, \dots, 2r-2$ to the $2r-3$ edges. Since the set $\{2, 3, \dots, 2r-2\}$ is congruent (mod r) to the set $\{1, 2, \dots, r\} \cup \{2, 3, \dots, r-2\}$, we will assign $1, 2, \dots, r$ to the r edges of the outer cycle in P_r^2 , and $2, 3, \dots, r-2$ to the $r-3$ edges of the inner path in P_r^2 .

S. Lo[19] proved that all odd cycles are edge-graceful, and we use such labeling on the outer cycle. We label $1, 2, 3, \dots, \frac{r-1}{2}, \frac{r+1}{2}, \frac{r+3}{2}, \frac{r+5}{2}, \dots, r$ to the edges $e_{1,2}, e_{2,4}, e_{4,6}, \dots, e_{r-3,r-1}, e_{r-1,r}, e_{r,r-2}, e_{r-2,r-4}, \dots, e_{3,1}$. Therefore the outer cycle is edge-graceful and the vertex sums of the vertices $v_1, v_2, v_4, \dots, v_{r-1}, v_r, v_{r-2}, v_{r-4}, \dots, v_3$ induced from this labeling are $1, 3, 5, \dots, 0, 2, 4, 6, \dots, r-1$.

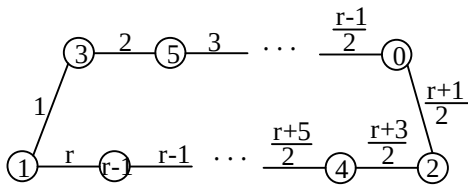


Figure 4: An edge labeling of the outer cycle

Then we assign $2, 3, \dots, r-2$ to the edges of the inner path in P_r^2 , by assigning $2, 4, \dots, r-3$ to the edges $e_{2,3}, e_{4,5}, \dots, e_{r-3,r-2}$, and assigning $3, 5, \dots, r-2$ to the edges $e_{r-2,r-1}, e_{r-4,r-3}, \dots, e_{3,4}$ respectively. Hence the vertex sums of vertices $v_2, v_3, v_4, v_5, \dots, v_{r-3}, v_{r-2}, v_{r-1}$ induced from this edge labeling are $2, 0, 2, 0, \dots, 2, 0, 3$. Notice that only the vertices $v_2, v_4, \dots, v_{r-3}, v_{r-1}$ have contributions from the edge labeling of the inner path.

Finally, we show that the vertex sums of the

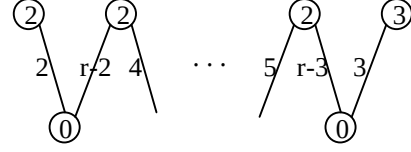


Figure 5: An edge labeling of the inner path

graph P_r^2 are pairwise distinct when associating the above labeling. We only consider the vertices $v_2, v_4, \dots, v_{r-3}, v_{r-1}$. The vertex sums of the vertices $v_2, v_4, \dots, v_{r-3}, v_{r-1}$ restricted in the inner cycle are $3, 5, \dots, r-2, 0$. And the vertex sum of the vertices $v_2, v_4, \dots, v_{r-3}, v_{r-1}$ restricted in the outer path are $2, 2, \dots, 2, 3$. Such the vertex sum of the vertices $v_2, v_4, \dots, v_{r-3}, v_{r-1}$ in P_r^2 are $5, 7, \dots, 0, 3$. Hence P_r^2 is 2-edge-graceful when r is odd, and we are done. $\square$

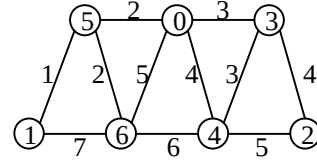


Figure 6: P_7^2 is 2-edge-graceful

Therefore we have shown the edge-graceful spectrum of P_r^2 , where r is odd and $3 \nmid r$. But if we would like to show the edge-graceful spectrum of P_r^2 , where r is odd and $3 \mid r$, we must prove in addition that P_r^2 is also $(n+2)$ -edge-graceful and $(2n+2)$ -edge-graceful, where $r = 3n$.

Theorem 4. P_r^2 is $(n+2)$ -edge-graceful, where $r = 3n$ is odd.

Proof. Similarly, let the path P_r is a graph with vertex set $\{v_1, v_2, \dots, v_r\}$ with the edge set $\{v_i v_{i+1} : 1 \leq i \leq r-1\}$. P_r^2 is a graph on the same set of vertices of P_r and where any pair of vertices of distance 2 or less are connected by an edge. Assume $e_{i,j}$ denote the edge $v_i v_j \in E(P_r^2)$. We partition the edges of the graph P_r^2 into the following sets: $A_1 = \{e_{1,2}, e_{1,3}, e_{2,3}, e_{2,4}, e_{3,4}, e_{4,5}\}$, $A_i = \{e_{j,j+2}, e_{j+1,j+3}, e_{j+2,j+3}, e_{j+2,j+4}, e_{j+3,j+4}, e_{j+4,j+5}\}$ where $j = 3(i-1)$ and i ranges from 2 to $n-1$, and $A_n = \{e_{r-3,r-1}, e_{r-2,r}, e_{r-1,r}\}$.

To show the graph P_r^2 is $(n+2)$ -edge-graceful, we must label $n+2, n+3, \dots, 7n-2$ to these $2r-3$ edges. Note that the set $\{n+2, n+3, \dots, 7n-2\}$ is congruent to the set $\{n-1, n, n+1\} \cup (\{1, 2, \dots, n-2\} \cup \{n+2, n+3, \dots, 3n\}) \cup (\{1, 2, \dots, n-2\} \cup \{n+2, n+3, \dots, 3n\})$. We need the following fact:

CLAIM 4.1. *The set $\{1, 2, \dots, n-2\} \cup \{n+2, n+3, \dots, 3n\}$ can be partitioned into $n-1$ disjoint subsets $a_i = \{x_i, y_i, z_i : x_i + y_i + z_i \equiv 0 \pmod{r}\}$, where $i = 1, 2, \dots, n-1$.*

We will prove the above claim 4.1 in the following Theorem 5. Denote $a_1 = \{x_1 = 3n, y_1, z_1\}$, $a_{n-1} = \{x_{n-1}, y_{n-1}, z_{n-1} = 2n\}$, and $a_n = \{x_n = n, y_n = n+1, z_n = n-1\}$. Then we will assign the integers of a_i to the edges of A_i , where i ranges from 1 to n , by labeling the integers $x_1, y_1, z_1, y_1, x_1, z_1$ to the edges $e_{1,2}, e_{1,3}, e_{2,3}, e_{2,4}, e_{3,4}, e_{4,5}$, the integers $x_i, y_i, z_i, y_i, x_i, z_i$ to the edges $e_{j,j+2}, e_{j+1,j+3}, e_{j+2,j+3}, e_{j+2,j+4}, e_{j+3,j+4}, e_{j+4,j+5}$ where $j = 3(i-1)$ and i ranges from 2 to $n-1$, and the integers x_n, y_n, z_n to the edges $e_{n-3,n-1}, e_{n-2,n}, e_{n-1,n}$, respectively.

Therefore the induced vertex sums $f^+ : V(P_r^2) \rightarrow Z_r$, defined by $f^+(u) = \sum \{f(u, v) : (u, v) \in E(P_r^2)\} \pmod{r}$, can be calculated as follows modulo r :

$$f^+(v_1) = x_1 + y_1 \equiv y_1$$

$$f^+(v_2) = x_1 + y_1 + z_1 \equiv 0 \equiv x_1$$

For i with $2 \leq i \leq n-1$:

$$f^+(v_{3(i-1)}) = (x_{i-1} + y_{i-1} + z_{i-1}) + x_i \equiv x_i$$

$$f^+(v_{3(i-1)+1}) = (x_{i-1} + y_{i-1} + z_{i-1}) + y_i \equiv y_i$$

$$f^+(v_{3(i-1)+2}) = (x_i + y_i + z_i) + z_{i-1} \equiv z_{i-1}$$

And the remaining vertex sums are:

$$f^+(v_{3n-3}) = (x_{n-1} + y_{n-1} + z_{n-1}) + x_n \equiv x_n$$

$$f^+(v_{3n-2}) = (x_{n-1} + y_{n-1} + z_{n-1}) + y_n \equiv y_n$$

$$f^+(v_{3n-1}) = (x_n + z_{n-1}) + z_n \equiv z_n$$

$$f^+(v_{3n}) = y_n + z_n \equiv 2n \equiv z_{n-1}$$

It is clearly that all the vertex sums are pairwise distinct, hence we are done. $\square$

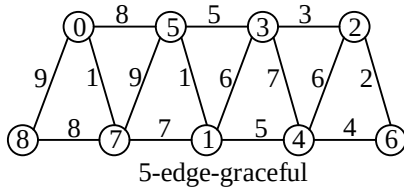


Figure 7: P_9^2 is 5-edge-graceful

While we prove the above theorem, how do we know that we can partition the set $\{1, 2, \dots, n-$

$2\} \cup \{n+2, n+3, \dots, 3n\}$ into the subsets $a_i = \{x_i, y_i, z_i : x_i + y_i + z_i \equiv 0 \pmod{r}\}$, where $i = 1, 2, \dots, n-1$? We show this fact below.

Theorem 5. *The set $\{1, 2, \dots, 3n\} - \{n-1, n, n+1\}$, where $n = 3, 5, \dots$, can be partitioned into $n-1$ 3-element subsets, and the sum of the elements in each subset is $0 \pmod{3n}$.*

Proof. We will give a constructive proof. If it can be shown that there is such partition for a set $\{1, 2, \dots, 3n\}$ which includes the set $\{n-1, n, n+1\}$, then we are done. Hence we divide into three cases as follows:

1. $n \equiv 0 \pmod{3}$.

Assign $S_i = \{3j-1, 3i, 3j+1 : (i \in \{1, 3, 5, \dots, n\}, i+2j = n) \text{ or } (i \in \{2, 4, 6, \dots, n-1\}, i+2j = 2n)\}$ with $-1 \equiv 3n-1 \pmod{3n}$. Also $S_{\frac{n}{3}} = \{n-1, n, n+1\}$.

2. $n \equiv 1 \pmod{3}$.

Assign $S_i = \{3i, 3j+1, 3j+2 : (i \in \{2, 4, 6, \dots, n-1\}, i+2j = n-1) \text{ or } (i \in \{1, 3, 5, \dots, n\}, i+2j = 2n-1)\}$. Also $S_{\frac{n+1}{3}} = \{n-1, n, n+1\}$.

3. $n \equiv 2 \pmod{3}$.

Assign $S_i = \{3j-2, 3j-1, 3i : (i \in \{2, 4, 6, \dots, n-1\}, i+2j = n+1) \text{ or } (i \in \{1, 3, 5, \dots, n\}, i+2j = 2n+1)\}$. Also $S_{\frac{n+2}{3}} = \{n-1, n, n+1\}$.

It is easy to see that $\bigcup S_i = \{1, 2, \dots, 3n\}$, $S_i \cap S_j = \emptyset$ for all $i \neq j$, and the sum of elements in each S_i is $0 \pmod{3n}$. $\square$

From the above, we know P_r^2 is $(n+2)$ -edge-graceful, where $r = 3n$ is odd. The $(2n+2)$ -edge-graceful labeling is basically obtained from shifting n to the $(n+2)$ -edge-graceful labeling. Let us show below how to get such $(2n+2)$ -edge-graceful labeling.

Theorem 6. *P_r^2 is $(2n+2)$ -edge-graceful, where $r = 3n$ is odd.*

Proof. Since P_r^2 is $(n+2)$ -edge-graceful, when r is odd and $r = 3n$. We have the disjoint sets $a_i = \{x_i, y_i, z_i : x_i + y_i + z_i \equiv 0 \pmod{r}\}$ where $i = 1, 2, \dots, n-1$. The $(2n+2)$ -edge-graceful labeling can be made by shifting the $(n+2)$ -edge-graceful labeling. Let the partite subsets be $a_i = \{x_i + n, y_i + n, z_i + n\}$, where $(x_i + n) + (y_i + n) + (z_i + n) \equiv 0 \pmod{r = 3n}$, for $i = 1, 2, \dots, n-1$, and $a_n = \{x_n = 2n, y_n = 2n+1, z_n = 2n-1\}$. Then we denote $a_1 = \{x_1 = 3n, y_1, z_1\}$, $a_{n-1} =$

$\{x_{n-1}, y_{n-1}, z_{n-1} = n\}$. Such assignment in a_i to the edges in A_i , when i ranges from 1 to n , is similar to the one in Theorem 4, hence it is $(2n + 2)$ -edge-graceful. $\square$

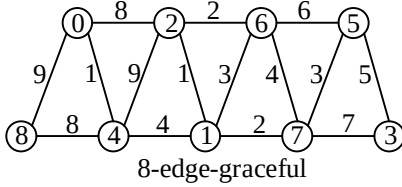


Figure 8: P_9^2 is 8-edge-graceful

By the above theorems, we completely determine the edge-graceful spectrum of graph P_r^2 for odd r . However the case for r a multiple of 4 still remains open. We give the labeling of first few special cases in the following section.

3 When the Number of Vertices is a Multiple of 4

We will consider the edge-graceful spectrum of P_r^2 where r is a multiple of 4, with special cases of few vertices.

Theorem 7. $egI(P_4^2) = \{1 + 2t : t = 0, 1, 2, 3, \dots\}$.

Proof. It suffices to show that P_4^2 is 1-edge-graceful and 3-edge-graceful by the lemma 1 and Theorem 2. Figure 9 shows that P_4^2 is 1-edge-graceful and 3-edge-graceful. And we have $1 + 4t$,

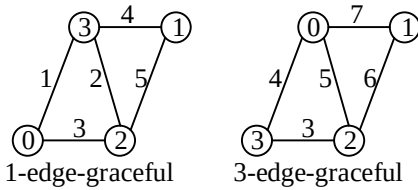


Figure 9: Edge-graceful-ness of P_4^2

$3 + 4t$ as the spectrum for all integers $t > 0$. Thus the edge-graceful-spectrum $egI(P_4^2) = \{1 + 2t : t = 0, 1, 2, 3, \dots\}$. $\square$

Theorem 8. $egI(P_8^2) = \mathbb{N}$, where $\mathbb{N}$ is the set of positive integers.

Proof. The graph P_8^2 has 8 vertices and 13 edges. By the necessary condition of being a k -edge-graceful graph, we have the equation $(12 + 2k)13 \equiv$

$\frac{8 \times 7}{2} \pmod{8} \Rightarrow 4 + 2k \equiv 4 \pmod{8} \Rightarrow k \equiv 0 \pmod{4}$. Again by the lemma 1, it suffices to show that P_8^2 is 0-edge-graceful and 4-edge-graceful, and Figure 10 provides these two labeling. $\square$

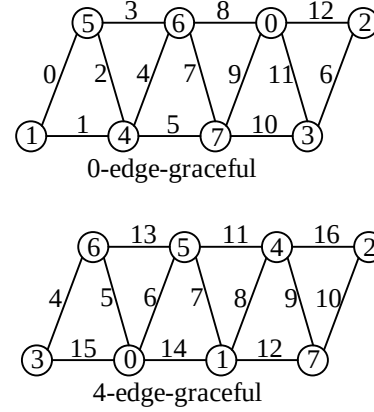


Figure 10: Edge-graceful-ness of P_8^2

Theorem 9. $egI(P_{12}^2) = \{2t + 1 : t = 0, 1, 2, \dots\}$.

Proof. Note that by the lemma 1 and Theorem 2, it suffices to show that P_{12}^2 is k -edge-graceful, where $k = 1, 3, 5, 7, 9, 11$. Figure 11 provides the required labeling. $\square$

The edge-graceful spectra of the first few P_r^2 are given, where r is a multiple of 4. However the general labeling for the square of paths is still not found. It is needed for completing the problem of determination of edge-graceful spectra of the square of paths.

4 Open Problems

In this paper we completely determine the edge-graceful spectrum of P_r^2 for odd r , while the problem of determining the edge-graceful spectrum of P_r^2 , where r a multiple of 4, remains open. Furthermore, the problem to determine the edge-graceful spectrum of higher power of paths P_r is also interesting and still unsolved.

In fact, the research of edge-graceful spectra of graphs provides insights for the problems of edge-graceful labeling. Along this direction, the edge-graceful tree conjecture or related problems can be studied for more details and hopefully be settled.

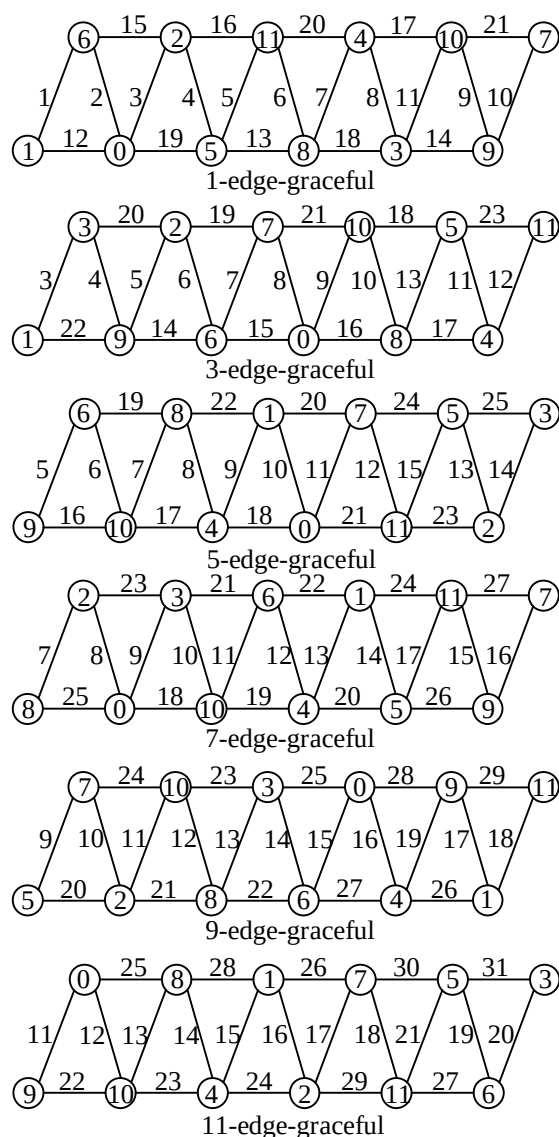


Figure 11: Edge-graceful-ness of P_{12}^2

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